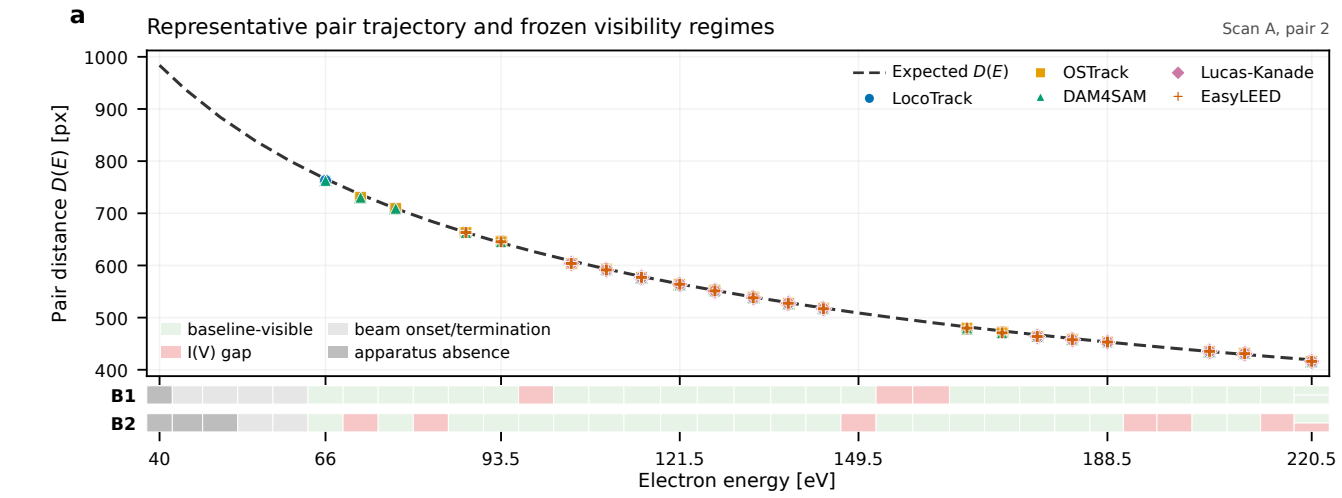

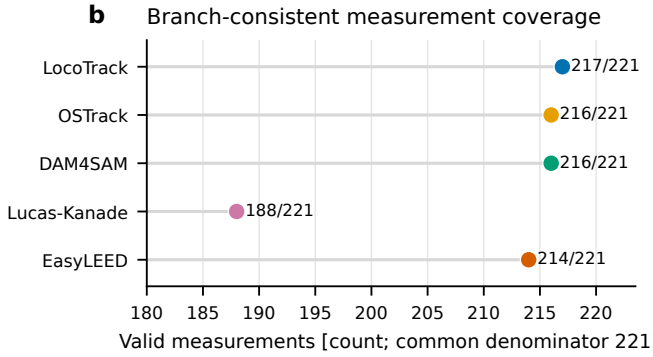

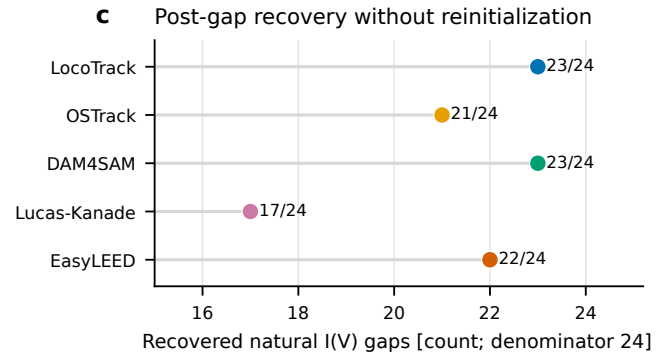

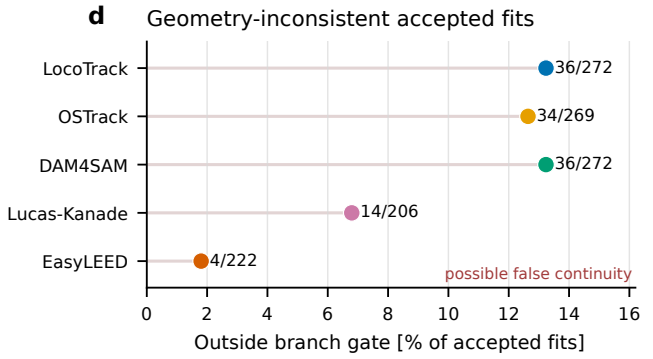

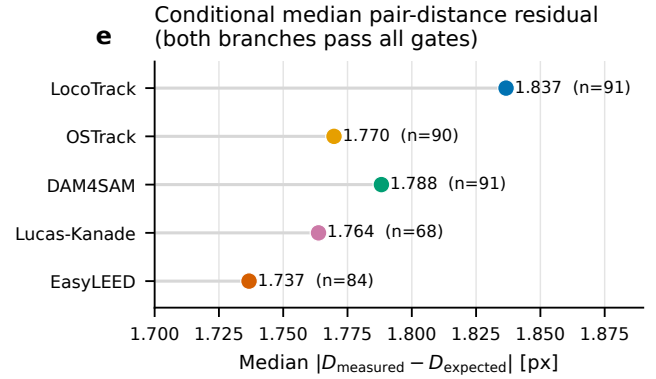

f External domain reference - separate initialization and denominator

Different native initialization

EasyLEED

one Kalman tracker per frozen spot

ViPErLEED

registered 18-beam whole pattern

Same downstream measurement

Valid Gaussian fits **38/60 both**

Gap checks **4/4 both**

Conditional median |D residual| [px]
Easy/VIP P1 0.462/0.460 (n=5 each)
Easy/VIP P2 0.762/0.760 (n=3 each)

Cross-method separation

0.626 px native proposals → common fit → **0.000986 px** Gaussian centers

maximum: 1.696 -> 0.0240 px

Restricted 15-image interval; ViPErLEED is not a sixth method and is not position ground truth.